

Practice Problems 1: System Modelling and Block Diagrams

Question 1

Neuromechanics is an area of biological research that studies how the nervous system controls the musculoskeletal system to bring about movement. A neuromechanics problem that the African Robotics Unit is interested in investigating is the role of the cheetah's tail in balancing its body. We have obtained footage from the Ann van Dyk Cheetah Centre of cheetahs being transported by bakkie. As the car bumps over the rough dirt roads around the sanctuary, the animals often lose their footing and are seen swinging their tails to recover.

a) Draw a block diagram modelling the cheetah in the vehicle. Here is the information your diagram must capture:

- The cheetah can be broken down into the following subsystems: the nervous system $N(s)$, the legs $L(s)$, the tail $T(s)$, the body $B(s)$, the eyes $E(s)$, the vestibular system $V(s)$ and the proprioceptive system $P(s)$.
- The cheetah is trying to control its body angle $\theta(s)$. It uses a combination of the eyes, vestibular system and proprioceptive system to detect this angle. You may consider the sensed angle to be the *average* of these three readings.
- The nervous system considers the difference between the desired angle $r(s)$ and the sensed position, and sends a nervous impulse $i(t)$ to the legs and tail to command them to respond.
- The legs and tail respond to the impulse, producing a combined torque $\tau(s)$ that acts to correct the body angle.
- The motion of the vehicle $c(s)$ also affects the angle of the body.

b) Reduce your block diagram into a transfer function relating the cheetah's body angle to (i) the desired angle, and (ii) the motion of the vehicle in terms of the given variables.

Question 2

Cape Town is a vitally important base for Antarctic research, as it is the closest port to the Antarctic circle. For this reason, marine robotics is one of the core research interests of the African Robotics Unit.

You wish to design a swarm of underwater robots to operate in the Southern ocean. To keep the robots working in a group, each one must be able to follow another robot at a set distance d .

a) Assuming the robots are moving in a straight line, use the information below to model the follower robot using a block diagram:

- The robot is propelled by an electric motor. Using a simplified model of the motor, we can relate the velocity of its shaft $\omega(t)$ to the input voltage $e(t)$ using the differential equation $J\dot{\omega}(t) = e(t) - B\omega(t)$, where J is the inertia of the motor and B is friction.
- The propellor converts the velocity of the motor to thrust. The thrust $F(t)$ can be assumed to be directly proportional to the shaft velocity, with proportionality constant ρ .
- The only two forces acting on the robot are the motor thrust, and hydrodynamic drag, which opposes the direction of motion. The robot has mass m . You can model the hydrodynamic drag as being proportional to the forward velocity $v(t)$, with proportionality constant D .
- The follower robot uses vision to determine the proximity to the leader, but this is not very reliable as murky water or poor lighting conditions can affect the result. You should assume the vision sensor *underestimates* the true distance by 10%.
- A controller applies a voltage to the motor based on the error between the desired distance d and the actual position $x(t)$. The controller is a PID, or *Proportional Integral Derivative* controller. Its output is a weighted sum of the error, its derivative and its time integral. The weights of these components are K , K_d and K_i respectively.
- Note: the output you are interested in is the position of the follower robot *relative* to the leader. Moving the follower forward will decrease this distance.

b) Based on your block diagram, devise the (i) open-loop and (ii) closed-loop transfer function for the robot in terms of the given variables and parameters.

Question 3

Feedforward control is a method where a known disturbance is predicted and preemptively counteracted. You want to use a combination of feedforward and feedback control to regulate the temperature in your tropical fishtank.

- You have a heater $H(s)$, a thermometer $T_1(s)$ that measures the temperature inside the tank $x(s)$, and a thermometer $T_2(s)$ that measures the ambient temperature of the room $a(s)$.
- A feedback controller $C(s)$ acts on the difference between the desired temperature $d(s)$ and the measured temperature in the tank.
- A feedforward controller $B(s)$ responds to the ambient temperature.
- The outputs of both controllers are combined to give the input to the heater.

Draw a block diagram of this system and derive transfer functions relating (i) the desired temperature and (ii) the ambient temperature to $x(t)$, in terms of the given variables.

Question 4

The Segway is a two-wheeled, self-balancing personal mobility device commonly used by security guards, tourists, people with trouble walking and Weird Al Yankovic. The Hoverboard is kind of the same thing, but smaller, and without a handle. (It doesn't hover.)

What these two devices have in common is their mechanism of operation: tilting the deck forward causes the wheels to accelerate forwards, while tilting backwards causes them to slow down.

This is how the speed is controlled, but also how balance is achieved: to accelerate the wheels forwards, a positive forward torque must be applied. From Newton's Third Law, this means that an equal and opposite torque is applied to the board, causing it to rotate back towards its original flat position. Likewise, applying a negative torque on the wheels will counteract backward rotation of the board.

Clearly this is a rocking and rolling example of feedback control! Let's model it.

To model the Segway/ hoverboard, create a block diagram and then reduce it to get a closed-loop transfer function relating the desired upright position $r(s)$ to the angle of the rider $\theta(s)$ in relation to the vertical axis.

Consider the following information:

- The rider and board can be modelled as an *inverted pendulum* – a common model system in control engineering. This is a nonlinear system, but it can be simplified to a linear model in the vicinity of the upright equilibrium point when the angle θ is close to zero, as at this point, the *small angle theorem* can be applied ($\sin(\theta) \approx \theta$).

Besides the torque due to gravity and the input torque τ applied by the motor, you may assume that friction creates a damping effect proportional to the angular velocity of the board, $-\mu\dot{\theta}$.

Devise a differential equation describing the motion of the inverted pendulum and hence, a transfer function for the rider block. Use the following symbols: m for the mass, J for the inertia, d for the distance from the pivot point (the board) to the centre of mass. (Assume initial rest.)

- The torque τ generated by the motor is proportional to the current i applied to its coil, with proportionality constant ρ .
- A gyroscopic sensor measures the *rate of change* of the angle i.e. $\dot{\theta}$



Figure 1: Weird Al on a Segway

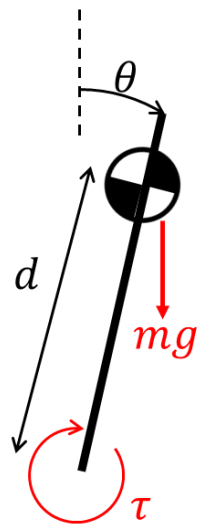


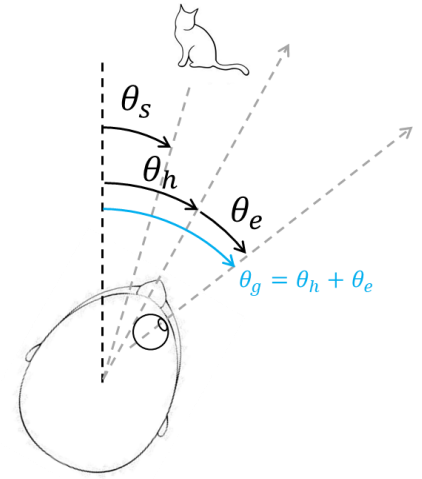
Figure 2: Inverted pendulum

- A microprocessor reads the angular rate from the gyroscope and implements *proportional derivative* control of the angle. Note that it will need to integrate the rate to obtain the proportional component of the error. The proportional gain is K_p and the derivative gain is K_d .

Question 5

In humans and many animals, visual tracking is a combination of movement of the head and eyes. This is beneficial as the eyes are able to rotate quickly with much less effort than the relatively large and heavy head, allowing the gaze to keep up with fast-moving subjects.

Draw a block diagram representing vision tracking based on the given information.



Hint: think of this control problem as having two loops: an inner loop controlling the position of the eyes relative to the head, and an outer loop controlling the position of the head. The difference between the gaze angle (sensed by the vision system, $V(s)$) and the angle to the subject causes the head to rotate towards the subject. Meanwhile, the eye will move within the head to make up the difference between the current head angle and the angle to the subject. The head sets the coarse direction of vision, while the eye makes the fine adjustments that allow gaze to lock onto the target.

- The gaze angle θ_g is a combination of the angle of the head θ_h and the angle of the eyes relative to the head θ_e . The angle to the subject θ_s is the target for the gaze angle.
- The desired angle for the eyes is the difference between the head angle (sensed by the vision system, $V(s)$), and the angle to the target. The eye angle is sensed by muscle spindles in the eye socket $S_e(s)$, and fed back to the neural controller of the eyes $C_e(s)$, which then sends impulses to the muscles of the eye socket $M(s)$. These produce a torque to rotate the eye $E(s)$.
- The nervous system sends impulses to the neck muscles $N(s)$, which produce torque to rotate the head $H(s)$. The neural controller of the head $C_h(s)$ responds to the difference between the gaze angle, sensed by $V(s)$, and the angle to the subject.

Based on your block diagram, devise a transfer function for visual tracking in terms of the given variables and subsystems.